



**YILDIZ TECHNICAL UNIVERSITY
FACULTY OF CHEMICAL AND METALLURGY
DEPARTMENT OF MATHEMATICAL ENGINEERING**

GRADUATION THESIS

**MULTI-OBJECTIVE OPTIMIZATION OF MULTI-
CLOUD RESOURCE ALLOCATION USING
MODERN PORTFOLIO THEORY: BALANCING
COST, AVAILABILITY, AND SUSTAINABILITY**

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LIST OF SYMBOLS

CO_2	Carbon Dioxide
CO_2e	Carbon Dioxide Equivalent
gCO_2e/kWh	Grams of CO_2 Equivalent per Kilowatt-Hour
$E(R_p)$	Expected Return of a Portfolio p
$F(w)$	Global Objective Function
$C(w)$	Portfolio Cost Function
$R(w)$	Portfolio Risk Function
$G(w)$	Portfolio Carbon Intensity Function
w_i	Workload Allocation Weight of Provider-Region Pair i
σ_p^2	Portfolio Risk
σ_{ij}	Covariance Between Provider-Region Pair i and j
σ_p^2	Total Variance (Risk) of a Portfolio
α, β, γ	Preference Coefficients / Objective Weights

LIST OF ABBREVIATIONS

API	Application Programming Interface
AWS	Amazon Web Services
AZURE	Microsoft Azure
GCP	Google Cloud Platform
MPT	Modern Portfolio Theory
QoS	Quality of Service
SLA	Service Level Agreement
SLSQP	Sequential Least Squares Programming

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PREFACE

This thesis was prepared as part of the requirements for the completion of the Bachelor's Degree in Mathematical Engineering at Yıldız Technical University. The inspiration for this research was deeply influenced by my professional tenure as an intern at Turkcell within the Cloud and Security Service Delivery department. While the technical scope of this study was developed independently, the opportunity to observe the operational intricacies of a leading telecommunications company provided a unique perspective on the critical challenges faced by modern digital enterprises. Witnessing the immense scale of infrastructure management and the high stakes of service reliability firsthand motivated me to explore how mathematical engineering principles could be utilized to create more robust, multi-cloud strategies.

Consequently, this study focuses on the multi-objective optimization of resource allocation in multi-cloud environments. By adapting Modern Portfolio Theory—a cornerstone of financial mathematics—this research treats cloud service providers as financial assets to identify an optimal balance between operational costs, system availability, and environmental sustainability. Throughout this project, I aimed to bridge the gap between theoretical modeling and practical infrastructure needs, specifically by utilizing covariance analysis to model service correlations and Pareto optimization to identify efficient resource distributions.

I would like to express my deepest gratitude to my supervisor, Prof. Dr. Ülkü BABUŞCU YEŞİL, for her invaluable guidance, constructive feedback, and continuous support throughout this study. I would also like to thank my family and friends for their constant encouragement, patience, and support throughout this process. This work represents both the culmination of my undergraduate education and a significant step toward applying advanced mathematical models to solve complex, real-world data problems.

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ABSTRACT

Cloud computing has become one of the main foundations of modern digital systems. However, depending on a single cloud provider may create several problems such as vendor lock-in, service interruption risk, and limited flexibility in infrastructure planning. For this reason, multi-cloud strategies have become increasingly important for organizations that need more resilient, cost-efficient, and sustainable cloud environments.

This thesis focused on the multi-cloud resource allocation problem by considering three main objectives at the same time: operational cost, SLA-related risk, and carbon intensity. In the proposed approach, the principles of Modern Portfolio Theory were adapted to cloud infrastructure planning. Each cloud provider-region pair was treated as an asset in a portfolio, and workload allocation ratios were represented as portfolio weights. The main idea was to distribute the workload across different cloud alternatives in a way that balances cost efficiency, service reliability, and environmental sustainability.

A Python-based simulation framework was developed to implement the proposed model. The framework included data preparation, normalization, covariance-based portfolio risk calculation, constrained optimization, and visualization steps. The optimization problem was solved by using the SLSQP algorithm under four different preference scenarios: cost-oriented, risk-oriented, green-oriented, and balanced. In addition, random feasible portfolios were generated to visualize the solution space and approximate non-dominated portfolios.

The experimental results showed that each scenario produced a different allocation behavior. The cost-oriented scenario achieved the lowest normalized cost score, but it also resulted in higher risk and carbon scores. The risk-oriented scenario achieved the lowest risk score by distributing the workload more broadly across provider-region pairs. The green-oriented scenario minimized the carbon score by assigning the workload mainly to EU-West regions. The balanced scenario did not minimize a single objective completely, but it provided a practical compromise between cost, risk, and sustainability.

Overall, the results showed that multi-cloud resource allocation should not be evaluated only from a cost perspective. A portfolio-based multi-objective optimization model can help decision-makers compare different allocation strategies more clearly and understand the trade-offs between operational cost, reliability, and sustainability. Although the implementation was developed as a simulation-based framework with scenario-based risk parameters, it provides a useful foundation for future studies on risk-aware and sustainability-aware cloud resource allocation.

Keywords: Multi-cloud, resource allocation, Modern Portfolio Theory, multi-objective optimization, SLA risk, carbon intensity, sustainability, cloud computing.

ÖZET

Bulut bilişim, modern dijital sistemlerin en önemli altyapı bileşenlerinden biri haline gelmiştir. Ancak yalnızca tek bir bulut sağlayıcısına bağlı kalmak, sağlayıcı bağımlılığı, hizmet kesintisi riski ve altyapı planlamasında esneklik kaybı gibi çeşitli problemlere yol açabilmektedir. Bu nedenle çoklu bulut stratejileri, daha dayanıklı, maliyet açısından daha verimli ve sürdürülebilir bulut altyapıları oluşturmak isteyen kurumlar için giderek daha önemli hale gelmektedir.

Bu tez çalışmasında, çoklu bulut kaynak tahsisi problemi üç temel amaç dikkate alınarak incelenmiştir: operasyonel maliyet, SLA ile ilişkili risk ve karbon yoğunluğu. Önerilen yaklaşımda, Modern Portföy Teorisi'nin temel prensipleri bulut altyapısı planlamasına uyarlanmıştır. Her bulut sağlayıcı-bölge çifti bir portföy varlığı olarak ele alınmış, iş yükü dağıtım oranları ise portföy ağırlıkları olarak temsil edilmiştir. Bu yaklaşımın temel amacı, iş yükünü farklı bulut alternatifleri arasında maliyet, güvenilirlik ve çevresel sürdürülebilirlik dengesini sağlayacak şekilde dağıtmaktır.

Önerilen modeli uygulamak için Python tabanlı bir simülasyon çerçevesi geliştirilmiştir. Bu çerçeve; veri hazırlama, normalizasyon, kovaryans tabanlı portföy riski hesaplama, kısıtlı optimizasyon ve görselleştirme adımlarından oluşmaktadır. Optimizasyon problemi SLSQP algoritması kullanılarak dört farklı tercih senaryosu altında çözülmüştür: maliyet odaklı, risk odaklı, çevreci ve dengeli senaryo. Ayrıca, çözüm uzayını ve yaklaşık Pareto-optimal portföyleri görselleştirmek amacıyla rastgele uygun portföyler üretilmiştir.

Elde edilen deneysel sonuçlar, her senaryonun farklı bir kaynak tahsisi davranışı oluşturduğunu göstermiştir. Maliyet odaklı senaryo en düşük normalize maliyet skorunu elde etmiş, ancak daha yüksek risk ve karbon skorlarına yol açmıştır. Risk odaklı senaryo, iş yükünü sağlayıcı-bölge çiftleri arasında daha geniş şekilde dağıtarak en düşük risk skoruna ulaşmıştır. Çevreci senaryo, iş yükünü ağırlıklı olarak EU-West bölgelerine yönlendirerek karbon skorunu minimum seviyeye indirmiştir. Dengeli senaryo ise tek bir amacı tamamen minimize etmemesine rağmen maliyet, risk ve sürdürülebilirlik arasında daha pratik bir denge sağlamıştır.

Genel olarak sonuçlar, çoklu bulut kaynak tahsisi probleminin yalnızca maliyet açısından değerlendirilmemesi gerektiğini göstermektedir. Portföy tabanlı çok amaçlı optimizasyon modeli, karar vericilerin farklı bulut tahsis stratejilerini daha açık şekilde karşılaştırmasına ve operasyonel maliyet, güvenilirlik ve sürdürülebilirlik arasındaki ödünleşimleri daha iyi anlamasına yardımcı olabilir. Bu çalışma senaryo tabanlı risk parametreleri kullanan bir simülasyon çerçevesi olarak geliştirilmiş olsa da, risk duyarlı ve sürdürülebilirlik odaklı bulut kaynak tahsisi üzerine yapılacak gelecek çalışmalar için faydalı bir temel sunmaktadır.

Anahtar Kelimeler: Çoklu bulut, kaynak tahsisi, Modern Portföy Teorisi, çok amaçlı optimizasyon, SLA riski, karbon yoğunluğu, sürdürülebilirlik, bulut bilişim.

1. INTRODUCTION

This chapter introduced the general framework of the thesis and outlined the main motivation, scope, and methodological direction of the study. First, the problem of resource allocation in multi-cloud environments was defined and its practical significance was emphasized. Then, the fundamental concepts required to understand the proposed approach were presented, followed by a brief evaluation of previous studies and the identification of the research gap. Finally, the methodological rationale of the thesis was explained and the overall structure of the study was summarized.

1.1. Definition and Significance of the Study

Cloud computing has become an indispensable infrastructure for modern digital enterprises, offering scalable resources on demand. However, reliance on a single cloud service provider introduces significant risks, such as vendor lock-in and catastrophic service outages that can render global platforms inaccessible. This study examined the optimization of resource allocation in a multi-cloud environment, where workloads were distributed across multiple providers to enhance system resilience. The significance of this research lies in its integrated approach to balancing three often-conflicting objectives: minimizing operational costs, maximizing service availability (uptime), and reducing the corporate carbon footprint. The scope of this study was limited to the allocation of compute resources across three major cloud service providers: Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). The analysis was based on provider-region pairs and focused on static allocation decisions, where workload proportions were determined for a given planning horizon rather than being adjusted dynamically in real time. In this context, the study considered cost, availability, and carbon intensity metrics as the primary criteria for evaluating alternative multi-cloud allocation strategies [1].

1.2. Preliminary Information

To understand the proposed optimization framework, several key concepts must be established:

Service Level Agreements (SLA): These are formal contracts between providers and users that guarantee specific performance metrics, primarily "uptime". A violation occurs when the provider fails to meet these targets, often resulting in financial penalties or credits [2], [3], [4].

Modern Portfolio Theory (MPT): Developed by Harry Markowitz, this mathematical framework is used to construct a portfolio of assets such that the expected return is maximized for a given level of risk [5].

Sustainability in Computing: This involves monitoring the "carbon intensity" of data centers, where different geographic regions and energy sources result in varying levels of CO₂

emissions for the same computational workload [6], [7], [8].

1.3. Research Gap and Contribution

Although cloud resource allocation has been widely studied in the literature, many existing approaches mainly focus on cost minimization, performance improvement, or quality-of-service-based scheduling. In addition, multi-cloud reliability is often evaluated by assuming that cloud providers operate independently. However, in practical cloud environments, service disruptions may be correlated due to shared regional infrastructure, network dependencies, or common geographical exposure.

The main contribution of this thesis is to propose a portfolio-based multi-objective optimization framework for multi-cloud resource allocation. By adapting Modern Portfolio Theory to cloud infrastructure planning, the study models provider-region pairs as portfolio assets and evaluates workload allocation decisions in terms of cost, SLA-related risk, and carbon intensity. In this way, the thesis aims to provide a structured decision-support framework that considers not only economic efficiency, but also service reliability and environmental sustainability.

1.4. Methodology and Rationale

The primary objective of this study was to develop a mathematical optimization framework for resource allocation in multi-cloud environments by adapting Modern Portfolio Theory (MPT) to cloud computing. In this framework, cloud service providers were modeled as portfolio assets, and workload distribution decisions were represented as allocation weights. The model was designed to determine an allocation strategy that minimizes operational cost and environmental impact while reducing system risk represented through SLA-related risk indicators [5], [9].

To achieve this objective, the multi-cloud allocation problem was formulated as a multi-objective optimization problem in which cost, availability, and carbon intensity were treated as competing criteria. Covariance analysis was employed to capture interdependencies and correlated failure risks between cloud providers. In addition, carbon intensity metrics were incorporated into the model as a sustainability-related decision component [6], [7], [8], [10].

This methodology was chosen because the Markowitz Mean-Variance approach provided a mathematically consistent way to evaluate both expected performance and risk within a single framework. Unlike traditional optimization methods that often assume provider independence, the proposed approach was intended to account for shared infrastructure risks and conflicting objectives. As a result, the model was used to identify approximate Pareto-optimal allocation strategies and to provide a more comprehensive basis for multi-cloud decision-making under different organizational priorities [9], [11].

1.5. Structure of the Thesis

To facilitate a clear understanding of the research, the subsequent sections of this study were organized as follows:

- **Chapter 2** reviewed the related literature and provided the theoretical background of Modern Portfolio Theory and multi-objective optimization.
- **Chapter 3** outlined the data collection and preparation process, including cloud pricing metrics, SLA-related reliability indicators, scenario-based risk parameters, and regional carbon intensity values.
- **Chapter 4** described the development of the Python-based simulation tool and the algorithmic implementation.
- **Chapter 5** presented the experimental results, analyzing the "Efficient Frontier" across various scenarios.
- **Chapter 6** concluded the study with a summary of findings and suggestions for future research.

2. THEORETICAL BACKGROUND AND MATHEMATICAL MODELING

This chapter established the mathematical framework necessary to optimize cloud resource allocation through the lens of financial mathematics. The discussion began with a review of the existing literature in order to contextualize the research gap, followed by a detailed examination of Modern Portfolio Theory (MPT). Subsequently, the principles of multi-objective optimization and Pareto optimality were defined to address the trade-offs between cost, availability, and environmental sustainability.

2.1. Literature Review and Related Work

The optimization of cloud resource allocation has been extensively studied in the field of distributed systems for over a decade. Early research primarily focused on cost minimization and latency-aware scheduling in single-provider environments, where cloud resources were typically modeled as independent variables and optimization was performed using linear programming or heuristic-based approaches to satisfy basic performance requirements [1], [11].

As cloud ecosystems have grown in complexity, the limitations of single-provider strategies have become increasingly evident. In particular, the risks of vendor lock-in and single points of failure have been identified as critical concerns for large-scale digital enterprises. To address these challenges, multi-cloud architectures have emerged as a standard approach for improving system resilience and service continuity by distributing workloads across multiple providers [1]. Existing studies on multi-cloud scheduling have primarily focused on load balancing and Quality of Service (QoS)-driven optimization, often utilizing heuristic or evolutionary algorithms to manage the inherent complexity of the problem [11].

A significant body of work has also incorporated Service Level Agreement (SLA) metrics into optimization frameworks [12]. These approaches typically aim to minimize SLA violations or their associated financial penalties, thereby capturing the economic implications of service reliability. However, most existing models assume that cloud providers operate independently and fail to account for potential correlations between them [12]. In practice, providers may share underlying physical infrastructure such as regional power grids and network backbones, which can lead to correlated failures and synchronized outages.

More recently, advanced optimization techniques, including machine learning and hybrid metaheuristic methods, have been proposed to enhance multi-objective performance across metrics such as latency, energy consumption, and availability [11]. Despite these developments, current approaches generally lack a unified framework that simultaneously incorporates inter-provider correlations and environmental sustainability considerations.

Furthermore, although the application of Modern Portfolio Theory (MPT)—originally developed for financial asset management—has been explored in limited contexts within cloud computing, these studies typically focus on performance variance reduction and do not

integrate sustainability metrics into the optimization process [9]. Similarly, carbon-aware computing and green cloud strategies have gained attention in the literature; however, carbon intensity is often treated as an isolated objective rather than being embedded within a comprehensive risk-return framework [6], [10], [13].

Therefore, a clear research gap exists in the integration of covariance-based risk modeling with multi-objective optimization in multi-cloud environments [1], [6], [9], [12]. This thesis aims to address this gap by proposing a unified framework that adapts Modern Portfolio Theory to model correlated risks between cloud providers, while simultaneously optimizing cost, availability, and environmental sustainability through a mathematically consistent and practically applicable approach.

2.2. Foundations of Modern Portfolio Theory (MPT)

Modern Portfolio Theory, introduced by Harry Markowitz, provides a mathematical formulation for diversifying assets to maximize expected returns for a given level of risk [5]. In this framework, the expected return of a portfolio ($E[R_p]$) is calculated as the weighted sum of the individual assets' returns:

$$E[R_p] = \sum_{i=1}^n w_i E[R_i] \quad (2.1)$$

The total risk of the portfolio, expressed as variance (σ_p^2), is determined not only by the individual variances of the assets but also by the covariance between them:

$$\sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij} \quad (2.2)$$

In this study, the covariance term (σ_{ij}) is utilized to model the correlation between cloud providers sharing physical or regional infrastructure [9].

2.3. Multi-Objective Optimization Model

In the proposed framework, the multi-cloud resource allocation problem was formulated as a multi-objective optimization problem [11]. Let w_i denote the proportion of workload allocated to provider-region pair i , where each w_i represents a decision variable of the model. These allocation weights were constrained such that the total workload distribution sums to one, that is $\sum_{i=1}^n w_i = 1$, and no negative allocation was permitted, so $w_i \geq 0$ for all i .

The objective structure of the model was defined based on three competing criteria: cost, risk, and environmental impact. The cost component was represented as the weighted sum of provider-specific unit costs, reflecting the total operational expenditure associated with a

given allocation strategy. The risk component was modeled through the portfolio variance structure of Modern Portfolio Theory, where the covariance matrix was used to represent both individual provider uncertainty and the correlation between provider-region pairs [9]. In this study, risk was defined in relation to Service Level Agreement (SLA) violation behavior. Therefore, the risk component of the model refers specifically to service reliability risk, rather than broader financial risks such as exchange rate volatility, sudden provider pricing changes, or long-term cost uncertainty. For each provider-region pair, the SLA-related risk value was interpreted as a scenario-based indicator of possible service unreliability rather than as a directly observed outage probability. These SLA-related risk values were used as provider-specific risk indicators, while the overall portfolio risk was evaluated through the variance-covariance structure of the allocation model. In this way, risk was represented not only by the individual unreliability of each provider, but also by the extent to which failures may occur in a correlated manner across different providers or regions. The covariance structure of the model was constructed to represent the degree of joint variability between provider-region pairs. In practical terms, covariance was intended to capture the possibility that service disruptions observed in one provider or region may coincide with disruptions in another due to shared physical or geographical dependencies. Such dependencies may arise from regional connectivity problems, common geographical exposure, or infrastructure-level dependencies. Therefore, the covariance matrix was used as a simplified mechanism for representing possible correlated risk behavior within the portfolio-based optimization framework. The environmental component was expressed as the weighted sum of regional carbon intensity values, allowing the total carbon footprint of the allocation to be evaluated [6], [7], [8].

Since these objectives are inherently conflicting, the problem was treated as a multi-objective optimization task rather than a single-objective minimization problem. In this study, the trade-off structure was evaluated through a Pareto-optimality perspective [11]. Accordingly, alternative allocation strategies were compared in terms of their relative performance across the three objectives, and the efficient solution set was interpreted as the set of allocations for which no criterion could be improved without worsening at least one of the others. This formulation provided a mathematically consistent basis for analyzing the balance between operational cost, service reliability, and sustainability in multi-cloud planning. In this context, improving availability was represented indirectly by minimizing SLA-related portfolio risk. The detailed construction of the scenario-based correlation coefficients and the resulting covariance matrix is explained in Section 3.4.

2.4. Sustainability and Carbon Intensity Constraints

Environmental sustainability was integrated as a mathematical constraint by quantifying the **Carbon Intensity** (C_i) of specific data center regions [6], [7], [8]. The total carbon footprint of the portfolio was formulated as:

$$Total\ Carbon = \sum_{i=1}^n w_i C_i \quad (2.3)$$

This formulation was intended to ensure that the resulting allocation followed global sustainability standards by treating carbon emissions as an economic penalty or a primary optimization objective [7], [8].

3. DATA COLLECTION AND EMPIRICAL FRAMEWORK

This chapter described the systematic process of gathering and preparing the empirical data required for the optimization model. The methodology relied on a structured simulation dataset that combined public cloud pricing information, SLA-related reliability indicators, scenario-based risk parameters, and regional carbon intensity values. The following sections detail the data sources, the criteria for provider selection, and the preprocessing steps used to construct the normalized objective values and the simplified covariance matrix.

3.1. Cloud Provider Selection and Pricing Metrics

For the empirical validation of the model, the "Big Three" cloud service providers—Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP)—were selected due to their dominant market share and the availability of public pricing, SLA, and sustainability-related information [1]. Pricing data were gathered for standardized compute instances (e.g., general-purpose virtual machines) across multiple geographic regions, such as US-East, EU-West (Frankfurt), and Asia-Pacific. The cost variable was treated as a deterministic input representing the hourly expenditure required for a unit of computational resource [14], [15], [16].

3.2. Availability and SLA Reliability Data

The reliability component of the model was represented through SLA-related risk indicators. Official Service Level Agreement (SLA) commitments from major cloud providers were reviewed together with publicly documented outage studies and reliability-related literature [2], [3], [4], [12]. However, since high-resolution and provider-region-specific outage datasets are not always publicly available, this thesis did not directly estimate real outage probabilities from raw historical outage records. Instead, SLA-related risk values were defined as scenario-based parameters that reflect relative differences in expected service reliability across provider-region pairs.

In this framework, the SLA risk value was used as the main risk indicator of each provider-region pair. These values were not intended to represent exact real-world outage probabilities. Rather, they provided a consistent input structure for testing the proposed portfolio-based optimization model and evaluating how risk-aware allocation decisions change under different preference scenarios.

3.3. Environmental Sustainability and Carbon Intensity Data

To address the sustainability objective, regional carbon intensity values were defined by using benchmark values informed by the Cloud Carbon Footprint framework and Electricity Maps methodology [7], [8]. These sources provide methodological guidance for estimating grams of CO₂ equivalent per kilowatt-hour (gCO₂e/kWh) based on the local electricity mix of a region [7], [8]. By mapping these intensity scores to the specific regions of AWS, Azure,

and GCP, the model incorporated the environmental cost of computation. This allowed the optimization algorithm to differentiate between regions with relatively higher and lower carbon intensity values [6], [7], [8].

3.4. Data Preprocessing and Covariance Construction

The input values gathered from public sources and scenario assumptions were subjected to a normalization process to ensure compatibility within the multi-objective functions. The most important preprocessing step was the construction of a simplified covariance matrix. Since detailed provider-region-specific outage datasets were not directly available, the covariance structure was constructed by using SLA-related risk values together with region-based correlation assumptions.

In this structure, provider-region pairs located in the same geographical region were assumed to have a higher correlation because they may be affected by similar regional infrastructure or connectivity risks. Provider-region pairs located in different regions were treated as less correlated within the scope of the simulation. In order to construct the covariance matrix in a mathematically consistent way, the scenario-based SLA risk values were first interpreted as relative risk indicators for each provider-region pair. Let r_i denote the SLA-related risk indicator of provider-region pair i , and let ρ_{ij} denote the assumed correlation coefficient between provider-region pairs i and j . The covariance between two provider-region pairs was calculated as follows:

$$\sigma_{ij} = \rho_{ij} \times r_i \times r_j \quad (3.1)$$

where σ_{ij} represents the covariance term used in the portfolio risk calculation. This formulation is based on the standard covariance identity used in portfolio theory, where covariance is expressed as the product of the correlation coefficient and the risk levels of two assets. Therefore, the covariance values were not assigned directly or arbitrarily; instead, they were derived from individual SLA-related risk indicators and a structured scenario-based correlation matrix.

Since detailed historical outage data at provider-region level were not publicly available, the correlation coefficients were selected as controlled scenario assumptions rather than empirically estimated parameters. The selection logic was based on the expected degree of shared infrastructure exposure. Provider-region pairs located in the same geographical region but belonging to different cloud providers were assigned a relatively high positive correlation, because they may be affected by common regional factors such as power grid failures, network backbone disruptions, natural disasters, or regional connectivity problems. Provider-region pairs belonging to the same provider but located in different geographical regions were assigned a moderate positive correlation, because provider-level dependencies, global control-plane incidents, or software-related failures may still create partial dependency across regions. Provider-region pairs belonging to different providers and different geographical regions were assigned a low positive correlation, reflecting the

assumption that they are mostly independent but not completely unrelated in a global cloud ecosystem.

The scenario-based correlation structure used in the simulation was defined as follows:

Table 3.1. Scenario-based correlation coefficient assumptions.

Relationship between provider-region pairs	Assumed correlation coefficient	Interpretation
Same provider-region pair	1.00	Perfect self-correlation on the diagonal of the correlation matrix
Different providers in the same region	0.60	Strong positive correlation due to shared regional infrastructure exposure
Same provider in different regions	0.30	Moderate positive correlation due to provider-level operational dependencies
Different providers in different regions	0.10	Weak positive correlation due to limited shared dependency

These coefficients were chosen to preserve the logical ordering of dependency strength: same-region exposure was assumed to create stronger correlation than same-provider cross-region exposure, while different-provider and different-region pairs were assumed to have the weakest dependency. All coefficients were kept within the valid correlation interval $[-1, 1]$, and the diagonal elements were set to 1.00. In this way, higher correlation coefficients increased the covariance contribution of provider-region pairs that were assumed to be exposed to common risks, whereas lower coefficients reduced the covariance penalty for geographically and operationally diversified allocations. This structure allowed the simulation to test the effect of correlated infrastructure risk without claiming that the selected coefficients represented exact real-world outage correlations.

This preprocessing stage transformed cost, risk, and carbon intensity inputs into a unified mathematical structure and enabled the Python-based simulation tool to evaluate the risk-adjusted performance of different multi-cloud resource allocation portfolios

4. ALGORITHMIC IMPLEMENTATION AND SIMULATION FRAMEWORK

This chapter details the development of the Python-based simulation tool designed to implement the multi-objective optimization framework described in the theoretical modeling phase. The software architecture was constructed to integrate cloud-related simulation inputs, including cost, SLA-related risk, and carbon intensity, into a mathematically consistent environment capable of identifying approximate Pareto-optimal resource distributions.

4.1. Development Environment and Software Architecture

The simulation engine was implemented using the Python programming language, chosen for its robust scientific computing ecosystem and suitability for complex mathematical simulations. The core architecture relies on several specialized libraries:

- NumPy: Utilized for high-performance matrix operations and the construction of the covariance matrix to model regional dependencies.
- SciPy (Optimize): The central component of the solver, employing the Sequential Least Squares Programming (SLSQP) algorithm to handle constrained minimization.
- Pandas: Used for data ingestion, preprocessing, and the management of multi-dimensional cloud datasets including pricing and carbon metrics.
- Matplotlib: Employed to visualize the "Efficient Frontier" and the distribution of feasible portfolios.

The tool was structured as a modular pipeline consisting of a Data Loader for simulation input preparation, a Portfolio Engine for mathematical modeling, an Optimizer for executing scenario-based simulations, and a Visualization component for analyzing feasible portfolios and approximate Pareto-optimal solutions.

4.2. Data Preprocessing and Feature Scaling

As established in the empirical framework, cloud metrics operate on different scales—hourly costs range from \$0.096 to \$0.120, while carbon intensity spans 200 to 600 gCO_2e/kWh . To prevent objective dominance by higher-magnitude variables, a **Min-Max Normalization** process was implemented within the simulation code. Each metric x was transformed into a dimensionless score x_{norm} between 0 and 1 using the formula:

$$x_{norm} = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (4.1)$$

This normalization allows the objective function to treat "Normalized Operational Cost," "Normalized Portfolio Risk (SLA Violation)," and "Normalized Carbon Intensity" as comparable scalar values within the weighted sum framework.

4.3. Implementation of the Multi-Objective Solver

The optimization problem was formulated as a constrained minimization task. The goal was to find a set of weights w (where w_i represents the workload proportion for provider-region pair i) that minimizes the global objective function $F(w)$:

$$F(w) = \alpha \cdot \text{Cost}(w) + \beta \cdot \text{Risk}(w) + \gamma \cdot \text{Sustainability}(w) \quad (4.2)$$

4.3.1. Dynamic Preference Weights

The simulation was designed to be dynamic, allowing for the adjustment of preference coefficients (α, β, γ) to reflect different organizational priorities:

- **α (Cost Coefficient):** Prioritizes operational expenditure minimization.
- **β (Risk Coefficient):** Prioritizes system reliability and SLA compliance through portfolio diversification and covariance analysis.
- **γ (Sustainability Coefficient):** Prioritizes the reduction of the carbon intensity score based on regional carbon intensity.

The solver strictly enforces the mathematical constraint that the sum of all coefficients must equal unity:

$$\alpha + \beta + \gamma = 1.0 \quad (4.3)$$

4.3.2. Constraints and SLSQP Algorithm

The **SLSQP (Sequential Least Squares Programming)** algorithm was selected due to its efficiency in solving non-linear optimization problems with both equality and inequality constraints. In this thesis, SLSQP was preferred over Pareto-focused evolutionary algorithms such as NSGA-II or MOEA/D for three main reasons. First, the proposed model was formulated as a scalarized multi-objective optimization problem by using the Weighted Sum Method. Therefore, the aim was not to generate the entire Pareto frontier directly, but to obtain optimal allocation solutions under predefined organizational preference scenarios. Second, the decision space was relatively small, since the simulation included a limited number of provider-region pairs and simple allocation constraints. Under these conditions, a gradient-based constrained optimization algorithm such as SLSQP was computationally efficient and suitable for obtaining deterministic scenario-based solutions. Third, SLSQP allowed the equality constraint on the sum of workload allocation weights and the non-negativity constraints to be handled directly within the optimization process.

For these reasons, SLSQP was considered appropriate for the scope of this study. However, it should be noted that this choice limits the optimization process to scenario-based scalar solutions rather than producing a complete Pareto front. Evolutionary multi-objective algorithms such as NSGA-II or MOEA/D would be more suitable if the main objective were to approximate the full Pareto frontier across all possible trade-off combinations. The primary constraints programmed into the solver were:

- **Sum of Weights constraint:**

$$\sum w_i = 1.0 \quad (4.4)$$

This constraint ensured 100% workload allocation across the selected providers.

- **Non-Negativity constraint:**

$$w_i \geq 0, i = 1, 2, \dots, n \quad (4.5)$$

This constraint prohibited negative resource allocation and short-selling of compute resources.

4.4. Experimental Scenario Definitions

To validate the model, four distinct operational scenarios were implemented by varying the preference weights (α, β, γ) . These scenarios demonstrate how the algorithm shifts resource allocation based on competing objectives:

- **Cost-Oriented Scenario** ($\alpha = 0.8, \beta = 0.1, \gamma = 0.1$): Focused on maximizing economic efficiency by allocating workloads to regions with the lowest unit costs.
- **Risk-Oriented Scenario** ($\alpha = 0.1, \beta = 0.8, \gamma = 0.1$): Focused on reducing the probability of synchronized outages via MPT-based diversification and covariance modeling.
- **Green-Oriented Scenario** ($\alpha = 0.1, \beta = 0.1, \gamma = 0.8$): Focused on minimizing environmental impact by selecting data centers with the lowest carbon intensity scores.
- **Balanced Scenario** ($\alpha = 0.33, \beta = 0.33, \gamma = 0.34$): An egalitarian approach seeking a compromise across cost, resilience, and sustainability.

4.5. Visualization and Efficient Frontier Construction

The final component of the simulation tool is the visualization engine. As shown in the generated simulation plots (e.g., *Efficient Frontier Analysis*), the tool executes thousands of random iterations to map the "Feasible Solution Cloud". By overlaying the scenario-based optimal points identified by the SLSQP solver, the tool visualizes the feasible portfolio space and the approximate Pareto-optimal portfolios. Since cost and risk are shown on the two-dimensional plane and carbon intensity is represented through the color scale, the Pareto-optimal solutions are interpreted as a projected approximation of the three-objective optimization space. This visualization provides a critical decision-support mechanism for identifying the optimal trade-offs between cost, risk, and sustainability in enterprise-scale infrastructure planning. In this study, the efficient frontier visualization should be interpreted as an approximate decision-support representation rather than a complete Pareto frontier generated by a dedicated multi-objective evolutionary algorithm. The random feasible portfolios were used to illustrate the trade-off structure of the solution space, while the SLSQP results represented the optimal allocations obtained under selected preference-weight scenarios.

5. RESULTS AND DISCUSSION

This chapter presents the experimental results obtained from the Python-based simulation framework developed in the previous chapter. The main objective of the analysis was to evaluate how different workload allocation strategies behaved under competing objectives, namely operational cost, SLA-related portfolio risk, and environmental sustainability. For this purpose, the simulation model generated a feasible portfolio space by producing 3000 random workload allocation portfolios and compared these feasible allocations with the optimal solutions obtained through the SLSQP algorithm. The results were evaluated under four different preference scenarios: cost-oriented, risk-oriented, green-oriented, and balanced.

5.1. Simulation Dataset and Scenario Outputs

The simulation was conducted using provider-region pairs created from three major cloud providers: AWS, Microsoft Azure, and Google Cloud Platform. Each provider was represented across three geographical regions: US-East, EU-West, and Asia-Pacific. For each provider-region pair, three numerical inputs were used: hourly operational cost, SLA violation risk, and regional carbon intensity. These values formed the basis of the multi-objective optimization model.

The first stage of the simulation integrated the cost, risk, and carbon intensity metrics into a unified dataset. An example of the processed dataset is shown in Table 5.1.

Table 5.1. Example of the processed simulation dataset.

Provider	Region	Hourly Cost	SLA Risk	Carbon Intensity
AWS	US-EAST	0.096	0.0010	450
AWS	EU-WEST	0.107	0.0008	200
AWS	ASIA-PACIFIC	0.120	0.0015	600
AZURE	US-EAST	0.096	0.0012	450
AZURE	EU-WEST	0.110	0.0009	200

Since the three variables were measured on different scales, Min-Max normalization was applied before the optimization process. This allowed operational cost, SLA-related risk, and

carbon intensity to be evaluated within the same weighted objective function. After preprocessing, the optimization model was executed under four scenarios by changing the preference coefficients α , β , and γ .

5.2. Scenario-Based Performance Scores

The normalized performance scores obtained from the optimization process are presented in Table 5.2. In this table, lower values indicate better performance for each objective because operational cost, portfolio risk, and carbon intensity were all formulated as minimization criteria.

Table 5.2. Scenario-based normalized performance scores.

Scenario	Cost Score	Risk Score	Carbon Score
Cost-Oriented	0.0000	0.4520	0.6250
Risk-Oriented	0.2845	0.2590	0.2624
Green-Oriented	0.3463	0.3594	0.0000
Balanced	0.3135	0.3284	0.0589

The cost-oriented scenario achieved the lowest possible normalized cost score, with a value of 0.0000. However, this scenario produced the highest risk score and the highest carbon score among the four scenarios. This result indicates that minimizing operational cost alone may lead to less reliable and less sustainable allocation decisions.

The risk-oriented scenario produced the lowest risk score, with a value of 0.2590. This result was consistent with the purpose of the scenario, since the optimization process prioritized service reliability and portfolio diversification. Unlike the cost-oriented scenario, the risk-oriented solution distributed the workload more broadly across different provider-region pairs, which reduced the concentration of risk.

The green-oriented scenario achieved the lowest carbon score, with a value of 0.0000. This means that the optimization algorithm successfully prioritized regions with lower carbon

intensity. However, the cost and risk scores of this scenario were higher than those of the balanced and risk-oriented solutions. Therefore, the green-oriented scenario demonstrated the trade-off between environmental sustainability and the other two objectives.

The balanced scenario provided a compromise between the three objectives. Although it did not produce the best individual score in any single category, it maintained relatively moderate values for cost, risk, and carbon intensity. This result showed that the balanced scenario can be interpreted as a practical allocation strategy when no single objective dominates the decision-making process.

5.3. Optimal Workload Allocation Results

The optimal workload allocation percentages obtained under each scenario are presented in Table 5.3. Each row represents a provider-region pair, while each column shows the percentage of workload assigned to that asset under a specific scenario.

Table 5.3. Scenario-based optimal workload allocation percentages.

Asset	Cost-Oriented	Risk-Oriented	Green-Oriented	Balanced
AWS_US-EAST	64.86%	19.22%	0.00%	6.12%
AWS_EU-WEST	0.00%	33.15%	74.24%	67.35%
AWS_ASIA-PACIFIC	0.00%	4.46%	0.00%	0.00%
AZURE_US-EAST	35.14%	10.04%	0.00%	3.31%
AZURE_EU-WEST	0.00%	19.47%	25.76%	23.22%
AZURE_ASIA-PACIFIC	0.00%	1.52%	0.00%	0.00%
GCP_US-EAST	0.00%	2.97%	0.00%	0.00%
GCP_EU-WEST	0.00%	9.06%	0.00%	0.00%
GCP_ASIA-PACIFIC	0.00%	0.12%	0.00%	0.00%

The cost-oriented scenario allocated the entire workload to US-East regions, with 64.86% assigned to AWS_US-EAST and 35.14% assigned to AZURE_US-EAST. This result was expected because these regions had the lowest operational cost values in the dataset. However, the concentration of the workload in a limited number of regions caused higher risk and carbon scores.

The risk-oriented scenario produced the most diversified allocation. Unlike the other scenarios, it assigned positive workload percentages to all provider-region pairs. The largest allocation was given to AWS_EU-WEST with 33.15%, followed by AZURE_EU-WEST with 19.47% and AWS_US-EAST with 19.22%. This distribution reflected the effect of the MPT-based covariance structure, since the algorithm reduced portfolio risk by spreading the workload across multiple assets.

The green-oriented scenario concentrated the workload in EU-West regions. Specifically, 74.24% of the workload was assigned to AWS_EU-WEST and 25.76% was assigned to AZURE_EU-WEST. This was consistent with the carbon intensity values used in the simulation, since EU-West had the lowest regional carbon intensity score. As a result, the green-oriented scenario achieved the minimum carbon score.

The balanced scenario also favored EU-West regions, allocating 67.35% of the workload to AWS_EU-WEST and 23.22% to AZURE_EU-WEST. In addition, smaller allocations were assigned to AWS_US-EAST and AZURE_US-EAST. This showed that the balanced scenario behaved similarly to the green-oriented scenario in terms of sustainability, but preserved some allocation to lower-cost US-East regions in order to reduce the cost penalty.

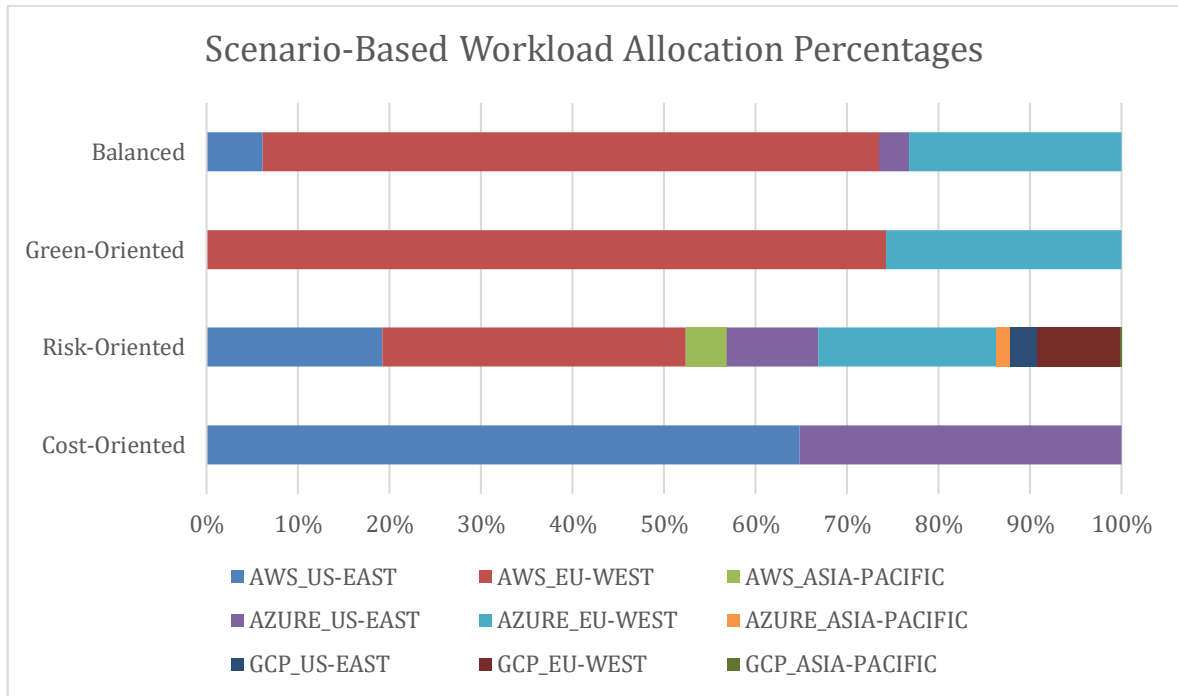


Figure 5.1. Scenario-based workload allocation

Figure 5.1 visualizes the workload allocation percentages obtained under the four optimization scenarios. The cost-oriented scenario concentrates the workload mainly in AWS_US-EAST and AZURE_US-EAST, whereas the risk-oriented scenario distributes the workload across a wider range of provider-region pairs. The green-oriented and balanced scenarios allocate most of the workload to EU-WEST regions, especially AWS_EU-WEST and AZURE_EU-WEST. This visualization supports the results presented in Table 5.3 and makes the scenario-based allocation behavior easier to interpret.

5.4. Feasible Portfolio Space and Approximate Non-Dominated Portfolios

The feasible portfolio space generated through random simulation is shown in Figure 5.2. In the figure, each point represents a different feasible workload allocation portfolio. The horizontal axis represents normalized operational cost, while the vertical axis represents normalized portfolio risk based on SLA violation behavior. The color scale represents normalized carbon intensity.

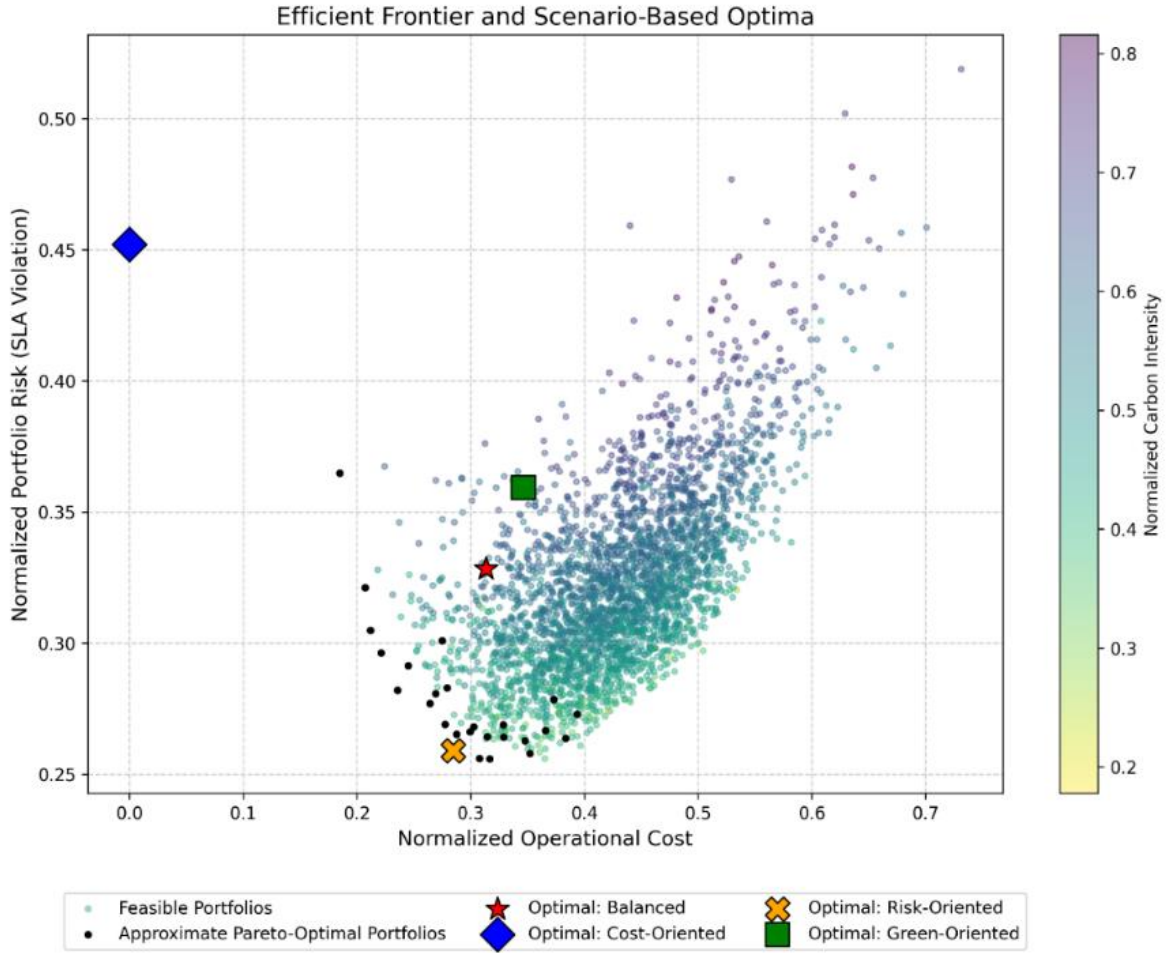


Figure 5.2. Feasible portfolio space and scenario-based optimal allocations

The black points in Figure 5.2 represent approximate non-dominated portfolios identified from the randomly generated feasible solution set by considering operational cost, SLA-related portfolio risk, and carbon intensity simultaneously. Since carbon intensity was represented through the color scale, these Pareto-optimal portfolios were projected onto the two-dimensional cost-risk plane. For this reason, some Pareto-optimal points may not appear optimal when only cost and risk are considered visually; however, they remain relevant in the three-objective optimization space because they provide advantages in terms of carbon intensity.

The feasible portfolio space showed that the optimization problem had a clear trade-off structure. Portfolios with very low cost did not necessarily provide low risk or low carbon intensity. Similarly, portfolios with lower carbon intensity were not always the cheapest alternatives. This result confirmed that multi-cloud resource allocation should not be evaluated using a single objective such as cost alone.

The scenario-based optimal points also supported this interpretation. The cost-oriented optimum was located at the lowest cost region of the plot but had a relatively high risk value. The risk-oriented optimum appeared close to the lower-risk region of the feasible solution space. The green-oriented and balanced solutions were located closer to the lower-carbon regions, which was consistent with their allocation of workload mainly to EU-West provider-region pairs.

5.5. Discussion of Trade-Offs

The results demonstrated that each scenario reflected a different organizational priority. The cost-oriented scenario was suitable for organizations that prioritize short-term operational expenditure reduction. However, its high risk and carbon scores suggested that cost minimization alone may create weaknesses in terms of resilience and sustainability.

The risk-oriented scenario provided the most reliable allocation in terms of normalized portfolio risk. Its diversified workload distribution reduced dependence on a small number of provider-region pairs. This result supported the main assumption of the thesis: Modern Portfolio Theory can be adapted to multi-cloud resource allocation by using diversification and covariance-aware risk modeling.

The green-oriented scenario showed that sustainability goals can strongly affect workload placement decisions. By assigning workload only to EU-West regions, the model minimized carbon intensity. However, this came at the expense of higher cost and risk scores compared to the risk-oriented and balanced scenarios. Therefore, carbon-aware allocation should be evaluated together with operational and reliability requirements.

The balanced scenario provided the most practical compromise among the four alternatives. It achieved a relatively low carbon score while keeping the cost and risk scores at moderate levels. Although it did not minimize any objective individually, it avoided the extreme weaknesses observed in the cost-oriented and green-oriented scenarios. For this reason, within the assumptions of this simulation, the balanced scenario can be considered the most suitable general-purpose allocation strategy among the tested alternatives.

Overall, the experimental results confirmed the usefulness of the proposed multi-objective optimization framework. The model made it possible to compare alternative cloud allocation strategies not only in terms of cost but also in terms of SLA-related risk and carbon intensity. This supports the idea that portfolio-based mathematical modeling can serve as a decision-support tool for enterprise-scale multi-cloud infrastructure planning.

6. CONCLUSIONS AND FUTURE WORK

This chapter concludes the thesis by summarizing the main findings obtained from the proposed multi-objective optimization framework. The study focused on the allocation of workloads across multiple cloud provider-region pairs by considering operational cost, SLA-related risk, and carbon intensity at the same time. In addition to the main conclusions, the limitations of the current implementation and possible directions for future research are also presented.

6.1. Conclusions

In this thesis, a multi-objective optimization framework was developed for multi-cloud resource allocation by adapting the principles of Modern Portfolio Theory. The main motivation of the study was based on the fact that cloud infrastructure decisions should not be made only according to operational cost. Although cost is an important factor for cloud users, service reliability and environmental sustainability have also become critical criteria in modern cloud infrastructure planning.

The proposed model treated each cloud provider-region pair as an asset in a portfolio. The workload allocation ratios were represented by portfolio weights, and the total workload was distributed under the constraint that the sum of all weights must be equal to one. In this way, the allocation problem was formulated as a constrained optimization problem. The model considered three main objectives: minimizing operational cost, minimizing SLA-related portfolio risk, and minimizing carbon intensity.

The mathematical structure of Modern Portfolio Theory was particularly useful for representing the risk dimension of the problem. Instead of evaluating cloud regions independently, the model included a covariance-based risk structure. This allowed the simulation to reflect the idea that provider-region pairs located in similar geographical regions may be affected by common regional infrastructure risks. Therefore, within the assumptions of the simulation, the model provided a more risk-aware allocation approach compared to a simple single-provider or single-objective selection strategy.

The Python-based implementation was developed as a modular simulation framework. The system included separate modules for data loading, portfolio metric calculation, constrained optimization, and visualization. Cost, SLA risk, and carbon intensity values were first normalized in order to make them comparable within the same objective function. Then, the optimization process was carried out by using the SLSQP algorithm under different preference scenarios.

The experimental results showed that each scenario produced a different allocation behavior. The cost-oriented scenario achieved the minimum normalized cost score. However, it also produced the highest risk and carbon scores. This result demonstrated that minimizing cost alone may lead to allocation decisions that are less reliable and less sustainable.

The risk-oriented scenario produced the lowest risk score among all scenarios. This scenario distributed the workload across all provider-region pairs and therefore represented the most diversified allocation strategy. This result supported the main assumption of the thesis that portfolio diversification can be used as a meaningful approach for reducing SLA-related risk in multi-cloud environments.

The green-oriented scenario achieved the minimum carbon score by allocating the workload entirely to EU-West provider-region pairs in the simulation. This result showed that carbon-aware optimization can significantly affect the final workload placement decision. However, this scenario also produced higher cost and risk scores compared to the balanced scenario, which confirmed the existence of a trade-off between sustainability and other operational objectives.

The balanced scenario provided a practical compromise between cost, risk, and sustainability. Although it did not minimize any objective individually, it avoided the extreme weaknesses observed in the other scenarios. It maintained moderate cost and risk scores while achieving a low carbon score. For this reason, the balanced scenario can be considered the most suitable general-purpose allocation strategy within the scope of this simulation.

Overall, the findings confirmed that multi-cloud resource allocation is a multi-objective decision problem. The results also showed that a portfolio-based mathematical framework can help decision-makers compare different allocation strategies more clearly. By evaluating cost, reliability, and sustainability together, the proposed model provides a structured decision-support approach for enterprise-scale cloud infrastructure planning.

6.2. Limitations of the Study

Although the proposed framework produced meaningful results, the study has several limitations. The first limitation is related to the data used in the simulation. The model was implemented by using a limited number of provider-region pairs from AWS, Microsoft Azure, and Google Cloud Platform. In a real enterprise environment, the number of available services, regions, instance types, and pricing models would be much larger.

The second limitation is related to the representation of SLA-related risk. Since high-resolution and provider-specific outage datasets are not always publicly available, SLA violation risk values were represented through scenario-based parameters. These values were useful for testing the proposed optimization framework, but they may not fully represent real-world outage behavior. In future studies, the model could be improved by using larger historical outage datasets or monitoring-based reliability data.

Another limitation is that the risk component of the proposed model was limited to SLA-related service reliability risk. Financial risks such as exchange rate fluctuations, sudden changes in cloud provider pricing, contract-based pricing differences, and long-term cost

uncertainty were not explicitly modeled. This limitation was mainly due to the scope of the study and the absence of a stable, provider-region-specific financial risk dataset that could be integrated into the same simulation framework. Therefore, the risk score used in this thesis should be interpreted as a service reliability-oriented risk measure rather than a comprehensive enterprise financial risk measure.

The third limitation is the simplified covariance structure. In the current implementation, provider-region pairs located in the same geographical region were assigned a positive correlation value, while cross-region dependencies were treated in a simplified manner.

In particular, the correlation coefficients used in the covariance matrix were defined as scenario-based assumptions according to regional and provider-level dependency logic, rather than being estimated from high-resolution historical outage data.

However, real cloud service failures may be affected by many different factors, such as shared network infrastructure, software dependencies, power supply conditions, and global service-level incidents. Therefore, a more detailed correlation model could improve the accuracy of the risk estimation.

Another limitation is related to the carbon intensity model. In this study, regional carbon intensity values were used as a proxy for environmental impact. However, the exact energy consumption of a cloud workload depends on several factors, including server utilization, cooling efficiency, hardware generation, data center energy contracts, and workload type. Therefore, the sustainability component of the model should be interpreted as an approximate environmental indicator rather than a complete carbon accounting system.

The model was also based on static workload allocation. This means that the optimization process produced allocation weights for a given planning scenario, but it did not dynamically update the allocation in real time according to changing demand, prices, carbon intensity, or service availability. In real cloud systems, dynamic resource scheduling may be required to respond to sudden workload changes or unexpected service interruptions.

Finally, the simulation focused mainly on compute-oriented resource allocation. Other cloud services such as storage, databases, networking, content delivery, and managed platform services were not included in the current scope. Including these components would make the model more comprehensive but also more complex.

6.3. Recommendations for Future Work

Future work can extend the proposed framework in several directions. First, the dataset can be expanded by including more cloud regions, more instance types, and additional cloud providers. This would allow the model to represent a wider range of real-world cloud infrastructure alternatives.

Second, the SLA-related risk model can be improved by integrating historical outage records, real-time monitoring data, or incident reports. With a richer reliability dataset, the covariance matrix could be estimated more accurately. This would make the risk component of the portfolio model more realistic and data-driven.

Third, the sustainability model can be extended by using dynamic carbon intensity data. Instead of assigning fixed regional carbon intensity values, future implementations could use time-dependent carbon intensity values from external data sources. This would allow the model to support carbon-aware scheduling decisions that change according to the electricity grid conditions of each region.

Fourth, the current static allocation model can be transformed into a dynamic optimization framework. In such a model, workload allocation weights could be updated periodically according to changes in price, demand, availability, and carbon intensity. This would make the framework more suitable for real-time cloud management systems.

Fifth, additional objectives can be added to the model. For example, latency, data transfer cost, compliance requirements, data residency restrictions, and performance variability can be included as additional decision criteria. These factors are especially important for enterprise systems that operate across different countries and regulatory environments.

Future studies can also extend the risk model by incorporating financial risk factors. For example, exchange rate volatility may affect cloud expenses for organizations that pay cloud bills in foreign currencies, while provider-side pricing changes may alter the expected cost of long-term workload allocation decisions. These factors can be represented through additional uncertainty parameters or scenario-based cost variance terms. In such an extended model, the risk function would not only capture SLA-related service unreliability, but also financial uncertainty associated with cloud pricing and currency fluctuations. This would make the framework more suitable for enterprise-level budgeting and long-term infrastructure planning.

Another possible improvement is the use of Pareto-focused multi-objective optimization algorithms. Although SLSQP was suitable for the weighted-sum and constraint-based structure of this thesis, it does not directly generate a complete Pareto frontier. Future studies could compare the proposed SLSQP-based framework with evolutionary multi-objective algorithms such as NSGA-II, MOEA/D, or other genetic algorithm-based approaches. These methods could approximate a wider set of non-dominated solutions and provide a more complete representation of the trade-off surface between cost, SLA-related risk, and carbon intensity. Such a comparison would also make it possible to evaluate whether the weighted-sum approach misses non-convex regions of the Pareto frontier.

Finally, the proposed model can be integrated into a decision-support dashboard. Such a system could allow users to change the preference coefficients interactively and observe how the optimal allocation changes under different business priorities. This would make the

framework more practical for cloud architects, infrastructure managers, and enterprise decision-makers.

In conclusion, this thesis demonstrated that Modern Portfolio Theory can be adapted to the multi-cloud resource allocation problem in a meaningful way. The proposed framework showed that cost, reliability, and sustainability should be evaluated together rather than separately. Although the implementation was developed as a simulation-based model, the results provided a strong foundation for future studies on risk-aware and sustainability-aware multi-cloud optimization.

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